

Article Summary

Brain tumor segmentation using deep learning: high performance with minimized MRI data

Huang et al. - *Frontiers in Radiology* (2025), 5:1616293

1. Overall Summary and Research Question

This retrospective study asks a practical medical-imaging question: can a deep-learning model segment gliomas accurately without requiring the conventional four-sequence MRI input (T1, T2, contrast-enhanced T1 [T1C], and FLAIR)? The authors trained a standard 3D U-Net and deliberately kept the network architecture fixed while varying only the MRI inputs. They compared T1C alone, FLAIR alone, T1C + FLAIR, and all four sequences. The target regions were enhancing tumor (ET) and tumor core (TC).

The central finding was that **T1C + FLAIR was the best overall reduced-input configuration**. On the independent test dataset it achieved median Dice scores of 0.867 for ET and 0.926 for TC, matching or exceeding the full four-sequence model (0.835 and 0.908). The study therefore argues that high-quality glioma segmentation can be achieved with substantially less MRI input data.

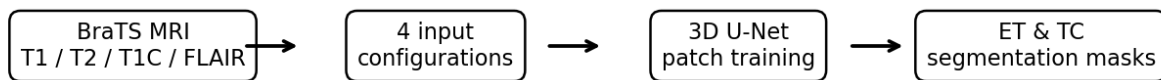


Figure 1. Simplified workflow of the study, reconstructed from the methods described by Huang et al.

2. Data Used

The study used benchmark datasets from the MICCAI Brain Tumor Segmentation (BraTS) challenges. Training used 285 glioma cases from BraTS 2018: 210 high-grade gliomas (HGGs) and 75 low-grade gliomas (LGGs). Testing used 358 separate cases: 66 from the held-out BraTS 2018 validation set and 292 from BraTS 2021. Cases missing MRI sequences in BraTS 2021 were excluded.

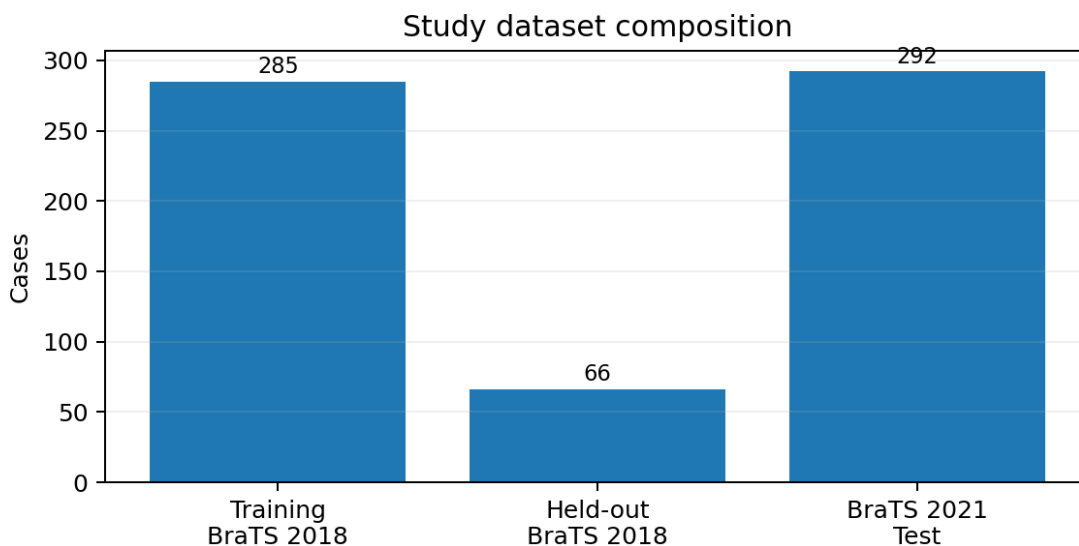


Figure 2. Dataset composition. The test cohort totaled 358 cases.

3. Preprocessing and Model

BraTS scans were already partially preprocessed and skull-stripped. Images were interpolated to 1 mm³ isotropic voxels and Z-normalized to zero mean and unit variance to reduce variation across institutions, scanners, and acquisition protocols. Data augmentation included rotations, translations, flipping, and intensity scaling; ground-truth masks were one-hot encoded.

The segmentation algorithm was a **3D U-Net**, an encoder-decoder convolutional neural network designed for volumetric biomedical segmentation. The model used four contraction/expansion levels, 3 x 3 x 3 kernels, BatchNorm, and 0.5 dropout. A final 1 x 1 x 1 convolution with softmax generated voxel-level probabilities. Separate models were trained for ET and TC, treating each voxel as tumor versus non-tumor for the selected subregion.

4. Training and Evaluation

Training was patch-based. Each sampled 3D patch measured 80 x 80 x 80 voxels, with 3,000 patches evaluated per epoch and a batch size of two. Patches containing fewer than 100 tumor voxels were discarded so that training would not be dominated by background. The model used Adam (learning rate 0.0001) and Dice-coefficient loss for 60 epochs. The implementation was designed to fit on a single NVIDIA GeForce RTX 2080 Ti GPU.

The authors performed 5-fold cross-validation on the 285-case training cohort, retrained models on the full training set, and then evaluated them on the 358-case test cohort. The primary metric was the **Dice score**, which measures spatial overlap between the predicted and reference segmentation masks. Secondary metrics were sensitivity, specificity, and 95% Hausdorff distance, the latter reflecting boundary disagreement.

5. Main Results

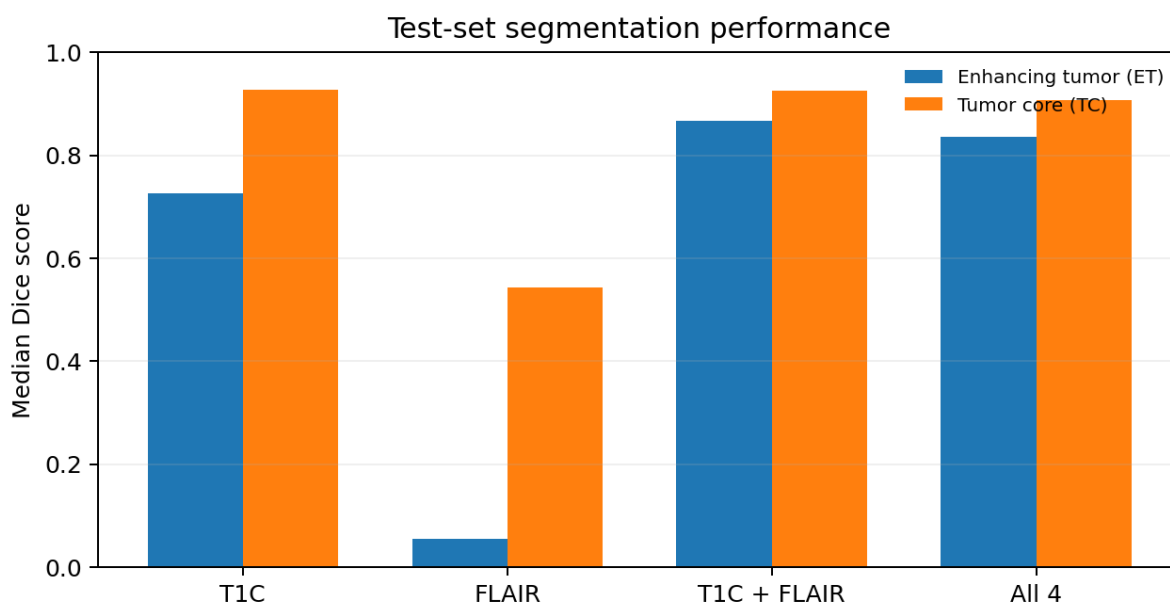


Figure 3. Median Dice scores on the independent test dataset. T1C + FLAIR produced the strongest balanced performance across ET and TC.

MRI input	CV ET	Test ET	CV TC	Test TC
T1C	0.781	0.726	0.852	0.928
FLAIR	0.008	0.056	0.619	0.543
T1C + FLAIR	0.814	0.867	0.856	0.926
T1 + T2 + T1C + FLAIR	0.785	0.835	0.841	0.908

T1C + FLAIR achieved the highest ET Dice score in both cross-validation (0.814) and testing (0.867). For TC, T1C-only (0.928) and T1C + FLAIR (0.926) were essentially comparable on the test set, both exceeding the four-sequence result (0.908). FLAIR alone performed poorly, especially for ET (test Dice 0.056), showing that FLAIR lacks sufficient information for reliable enhancing-tumor delineation by itself.

Statistically, there was no significant cross-validation difference among T1C, T1C + FLAIR, and the four-sequence configuration for either ET ($p = 0.33$) or TC ($p = 0.62$). Adding the poorly performing FLAIR-only group made the across-group differences significant (ET $p = 0.007$; TC $p = 0.01$).

6. Secondary Metrics and Computational Efficiency

For ET, T1C had the highest reported sensitivity (0.829), while T1C + FLAIR had the lowest 95% Hausdorff distance (5.964), indicating excellent boundary agreement. Specificity remained very high across all configurations (at least 0.958). For TC, the full four-sequence model had the highest sensitivity (0.754), but T1C and T1C + FLAIR remained strong overall.

Reducing MRI inputs also reduced computation. The reported time per epoch was approximately **7.5 minutes for one sequence, 9.5 minutes for two sequences, and 11 minutes for all four sequences**. T1C, T1C + FLAIR, and the four-sequence model generally reached strong performance before the planned 60 epochs, suggesting limited benefit from substantially longer training.

7. Interpretation and Clinical Significance

The result is clinically meaningful because full multiparametric MRI may be unavailable, costly, time-consuming, or difficult to standardize across institutions. The doublet combines complementary information: T1C emphasizes contrast-enhancing tumor, whereas FLAIR is sensitive to abnormal fluid-related signal and non-enhancing tumor-associated changes. Together, they supplied enough information for robust segmentation of both ET and TC in this dataset.

The findings also show that more imaging channels do not automatically produce a better deep-learning model. The four-sequence configuration imposed greater computational burden yet did not outperform the two-sequence model in the principal Dice-score comparisons. The paper therefore emphasizes **informative data selection** rather than simply maximizing input volume.

T1C alone was particularly effective for TC, but the authors note that low-grade gliomas may enhance weakly. Adding FLAIR helped compensate for this limitation, making T1C + FLAIR the most balanced choice across both HGG and LGG cases. FLAIR alone, by contrast, was ineffective for ET and inferior overall for the targeted subregions.

8. Strengths

A major strength is the controlled experimental design: the authors fixed the 3D U-Net architecture and changed the MRI sequence combinations, making the effect of input data easier to interpret. They also used 5-fold cross-validation followed by testing on a separate 358-case cohort. For 66 held-out BraTS 2018 cases, performance was evaluated through the MICCAI/CBICA portal against blinded ground truth, helping reduce evaluation bias. The use of established, expert-annotated BraTS datasets also strengthens reproducibility.

9. Limitations

The dataset does not represent the full anatomical diversity encountered in clinical neuroradiology. BraTS 2018 and 2021 contain supratentorial gliomas, so the approach was not evaluated for tumors in more difficult locations such as the posterior fossa or brainstem. The training cohort was also weighted toward HGGs (210 of 285 cases), and the authors acknowledge that performance of T1C-containing configurations might differ in a population with a larger proportion of LGGs.

The study also evaluates benchmark MRI rather than a prospective clinical workflow. Although the source data came from multiple institutions, scanners, and protocols, further external and prospective validation would be necessary before assuming equivalent performance in routine practice. The four-sequence ET model showed somewhat erratic training behavior, so it is possible that additional optimization could alter its performance, although this would increase computational cost.

10. Conclusion

Huang et al. conclude that high-performance 3D glioma segmentation does not necessarily require all four conventional BraTS MRI sequences. A **T1C + FLAIR doublet achieved consistently strong segmentation and matched or exceeded the four-sequence model** for the principal Dice-score comparisons. Reducing sequence dependency may lower data acquisition and computational requirements and make automated segmentation more practical in research, community, and resource-constrained environments.

Bottom line: This paper provides evidence that carefully chosen MRI sequences can matter more than simply providing a deep-learning model with every available sequence. For this 3D U-Net and BraTS glioma dataset, T1C + FLAIR offered the best balance of segmentation accuracy, computational efficiency, and reduced data requirements.

Source: Huang J, Yagmurlu B, Molleti P, et al. Brain tumor segmentation using deep learning: high performance with minimized MRI data. *Frontiers in Radiology*. 2025;5:1616293. doi:10.3389/fradi.2025.1616293.